Hardware and Circuit Guide

ESP32-CAM on a pan-tilt mount: bench circuit, flashing, tests and mounting

> **RULE:** Check every wire twice before connecting power. Wrong 5 V / 3.3 V connections can destroy the board. Ask Shreyas if unsure.

1. Parts checklist

- [ ] ESP32-CAM (AI-Thinker, with PSRAM and OV2640 camera) + ESP32-CAM-MB USB programmer board (or a 3.3 V FTDI adapter)
- [ ] Pan-tilt bracket + 2 micro servos (SG90 or MG90S)
- [ ] Separate 5 V power for the servos: 5 V, 2 A adapter or a buck converter (LM2596 / MP1584 set to 5.0 V)
- [ ] Capacitors: 470-1000 uF electrolytic (servo supply), 100 nF ceramic
- [ ] Jumper wires, breadboard, a small screwdriver, a multimeter
- [ ] A laptop with Arduino IDE (ESP32 board package) and Python 3
- [ ] Phone or tape measure for distance tests; a small tripod or clamp to hold the camera

2. Wiring (bench version)

From	To	Notes
ESP32-CAM 5V + GND	USB programmer board (or a stable 5 V, 1 A+ supply)	Camera streaming + WiFi pulls current; a weak USB port causes resets
Pan servo signal (orange/yellow)	ESP32-CAM GPIO 13	Pan = left/right
Tilt servo signal	ESP32-CAM GPIO 14	Tilt = up/down
Both servo red wires (+5 V)	SEPARATE 5 V supply (+)	Do NOT power servos from the ESP32-CAM 5 V pin
Both servo brown/black wires (GND)	Separate supply (-) AND ESP32-CAM GND	Grounds MUST be joined (common ground)
470-1000 uF capacitor	Across servo supply + and -	Long leg to +. Absorbs servo current spikes
GPIO 0 to GND (only when flashing)	Jumper, remove after flashing and press RESET	Skip this if you use the MB board (auto-flash)

Pin choice notes: the ESP32-CAM camera already uses most GPIOs. GPIO 13 and 14 are free when you do not use the SD card. Avoid GPIO 12 (boot-voltage strapping pin: servo signal can stop the board booting), GPIO 15, GPIO 2 and GPIO 0 for servos. GPIO 4 is the bright flash LED.

> **WATCH OUT:** Brownout is the number one problem: symptoms are 'Brownout detector was triggered' in Serial Monitor, random resets, or a frozen video. Fix: separate servo supply, common ground, big capacitor, short thick power wires, and a good USB cable. Our main rover had the same issue.

3. Flash the stock camera example first

1. Arduino IDE: Boards Manager, install 'esp32 by Espressif'. Select board **AI Thinker ESP32-CAM**.
2. File, Examples, ESP32, Camera, **CameraWebServer**. Uncomment `#define CAMERA_MODEL_AI_THINKER` and comment the others.
3. Put your WiFi name and password in the sketch (use a phone hotspot if needed; do not share these in the paper or GitHub).
4. Upload. Open Serial Monitor at 115200; note the IP address; open it in the browser; press Start Stream.
5. Take screenshots: the stream page and the Serial Monitor. Save as `evidence/hardware/`.
6. Test the snapshot URL `http://IP/capture`; it must return one JPEG. The Python capture script uses this.

4. Add the pan-tilt servos

Use `Starter Code/pan_tilt_snippet.ino` as the starting point, then ask an AI tool to add an HTTP handler like `/look?pan=60&tilt=100` (Prompt 1.3 in file 08). Two details that often cause jitter on ESP32-CAM: the camera clock already uses PWM timer 0, so give the servos other timers (the snippet does this); and always limit the angle range so the bracket does not hit its stops.

Test plan: command 5 pan angles and 5 tilt angles, each 3 times; measure the real angle with a phone protractor app or paper marks; note jitter and resets. Log each test in `Templates/pan_tilt_tests.csv`.

5. Image quality tests (these become a table in the paper)

OV2640 is a 2 MP camera with a manually focused lens. For leaf close-ups, twist the lens (carefully, small turns) until a leaf 10-25 cm away is sharp.

Test	Settings to vary	What to record (camera_tests.csv)
Resolution	VGA 640x480, SVGA 800x600, XGA 1024x768	Frame time (ms), file size (KB), sharpness by eye
JPEG quality	10, 12, 15 (lower number = better quality, larger file)	File size, visible artifacts
Distance	10, 20, 30, 50 cm from a leaf	Is the leaf sharp? Can you see lesions?
Lighting	Direct sun, shade, cloudy, indoor lamp	Over/under-exposure; tweak brightness / AEC
Stability	Run the stream 10 minutes	Resets, heat, WiFi drops

6. Mounting on the rover (when the bench works)

- Mount the camera 30-60 cm above the ground, looking slightly down at plant height.
- Shield the lens from direct sunlight glare; add a small hood.
- Strain-relieve the servo wires so panning does not tug connectors.
- Keep the camera power wiring away from motor wires (noise and current spikes).
- Take a clear photo of the finished mount from the front, side and top for the paper's Fig. (without faces or names).

7. What to hand over from this file

[ ] Photo of the bench circuit and a clean wiring diagram (hand-drawn photo is fine)

[ ] Screenshot of working stream and the Serial Monitor boot log

[ ] camera_tests.csv and pan_tilt_tests.csv filled in

[ ] Notes: what failed, what fixed it (brownout, jitter, focus). Honest troubleshooting notes make a good 'Limitations' paragraph